

An Analysis of Superposition in Continuous Chain-Of-Thought

Benjamin Roderick^{1,†} Edmand Yu^{1,†*} Yuchen Wu^{1,†}

Michael Rizvi-Martel^{2,‡}

¹McGill University ²Mila & Université de Montréal

[†]Student [‡]Mentor

Abstract

Recently, **chain-of-thought** (CoT) has evolved from discrete natural language prompting to **continuous CoT** methods that reason directly in embedding space. Techniques such as **Soft Thinking** utilize continuous concept tokens to encode multiple meanings simultaneously via superposition. However, the effectiveness of such methods hinges on the assumption that convex combinations in embedding space decode to coherent mixtures of distributions in probability space. In this work, we systematically evaluate this assumption. We conduct our investigation using three LLMs: DeepSeek-R1-Distill-Qwen-1.5B, DeepSeek-R1-Distill-Qwen-7B, and Qwen3-4B-Instruct-2507. By synthesizing vectors from varying grouping strategies, including semantic similarity, grammatical category, and random baselines and passing them through the model’s unembedding matrix, we analyze whether the resulting logits preserve a superposition of the original concepts. Our study seeks to identify the critical point, if any, where this interpretation breaks down. To the best of our knowledge, this is the first work to systematically test this foundational validity of continuous reasoning paradigms, offering a better understanding of the limitations and reliability of implicit reasoning mechanisms.

1 Introduction

Recently, **chain-of-thought** (CoT) has become the standard for reasoning in large language models (LLMs). Initially, CoT simply involved giving the model more "thinking time" by prompting it to produce intermediate reasoning steps during inference (Wei et al., 2022). Other methods, such as supervised fine-tuning (Yu et al., 2023) and reinforcement learning optimization (Shao et al., 2024), have also been explored. However, this form of reasoning remains constrained to the space of natural

language. To overcome this limitation, researchers began exploring models capable of reasoning without the use of explicit tokens. For instance, *Let’s Think Dot by Dot* (Pfau et al., 2024) uses special "thinking" tokens to facilitate internal reasoning. Another avenue is **continuous chain-of-thought** (Hao et al., 2024), which generates reasoning steps directly in the embedding space rather than in token space. Additionally, it has been shown that continuous CoT can also represent multiple potential reasoning paths simultaneously, enabling more efficient and expressive inference than discrete CoT through "superposition" of concepts (Zhu et al., 2025). A related, training-free method called **Soft Thinking** (Zhang et al., 2025) extends this idea by introducing *concept tokens*, which are continuous representations that encode multiple related meanings from discrete tokens. The authors report that **Soft Thinking** improves standard CoT while keeping outputs interpretable and readable. However, a fine-grained understanding of the success of this method remains elusive. The effectiveness of **Soft Thinking** hinges on the assumption that convex combinations in embedding space decode to a mixture of distributions in probability space. As noted by the authors (Zhang et al., 2025), using *concept tokens* at inference places the model in an out-of-distribution regime, since the model was never exposed to such continuous representations during training. This limitation motivates our primary research question: whether superpositions in embedding space correspond to superpositions in probability space. To the best of our knowledge, no prior work has systematically evaluated this assumption, and in this project, we aim to do so.

In this work, we investigate how word embeddings behave when combined under different grouping strategies. We conduct this investigation using three LLMs: DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI, 2025), DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), and Qwen3-4B-

Corresponding author.
edmand.yu@mail.mcgill.ca

Contact:

Instruct-2507 (Team, 2025a). Specifically, by sampling embeddings from varying grouping strategies, including semantic similarity, grammatical category, and random baselines, combining them via (non-)uniform convex combination, and passing the resulting vector through the model’s unembedding matrix, we analyze whether the resulting logits preserve a superposition of the original concepts. Our study seeks to determine if there is a critical point where this interpretation breaks down, thereby empirically testing the foundational validity of continuous reasoning paradigms like **Soft Thinking**.

The contribution of this work is that, to the best of our knowledge, no prior work has systematically evaluated the assumption that superpositions in embedding space correspond to superpositions in probability space. This study offers a better understanding of superposition in continuous chain-of-thought and highlights its limitations. Although this is a preliminary investigation regarding this assumption, we hope it motivates further research into the reliability and interpretability of these reasoning methods.

2 Related Work

Chain-of-Thought (CoT) Reasoning. CoT reasoning enhances the multi-step reasoning capabilities of large language models by introducing explicit intermediate steps. Common approaches include prompt-based learning methods (Wei et al., 2022), supervised fine-tuning (Yu et al., 2023), and reinforcement learning optimization (Shao et al., 2024). However, this form of reasoning remains constrained to the discrete space of natural language. In response, researchers have begun exploring models that can reason independently of explicit tokens, allowing the reasoning process to unfold apart from its linguistic forms. One promising direction is **Continuous Space Reasoning**.

Continuous Space Reasoning. Recently, significant attention has been directed toward architectures that operate within continuous vector spaces rather than discrete vocabularies. It has been shown that continuous CoT can represent multiple potential reasoning paths simultaneously, enabling more efficient and expressive inference than discrete CoT through "superposition" of concepts (Zhu et al., 2025). Prominent work in this direction includes **Chain of Continuous Thought** (Hao et al.,

2024), which generates reasoning steps directly in the embedding space, and a training-free approach **Soft Thinking** (Zhang et al., 2025). **Soft Thinking** introduces *concept tokens*, which are constructed by aggregating the vocabulary’s token embeddings based on their predicted probabilities, thereby defining a continuous concept space. Typically, CoT requires the model to sample one specific word at every step. This method eliminates this constraint by replacing that discrete sampling with continuous *concept tokens*, allowing the model to propagate multiple meanings simultaneously. This approach offers two significant innovations. First, by functioning within the continuous concept space defined as a convex combination of all token embeddings, the model is able to represent and manipulate both abstract concepts and fine-grained semantic information. Second, since each concept token maintains a probability distribution over all possible next tokens, the model can implicitly and efficiently explore multiple reasoning paths simultaneously, rather than being confined to a single trajectory. Moreover, the authors report that **Soft Thinking** improves standard CoT while keeping outputs interpretable and readable. However, the effectiveness of **Soft Thinking** hinges on the assumption that convex combinations in embedding space decode to a mixture of distributions in probability space. Therefore, in this work, we experimentally investigate whether superpositions in embedding space correspond to superpositions in probability space, which to the best of our knowledge, no prior work has systematically evaluated this assumption.

3 Data and Environment

Models. We select three widely used open-source LLMs to evaluate whether the superposition assumption holds across different model scales: DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI, 2025), Qwen3-4B-Instruct-2507 (Team, 2025a), and DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025). In addition, we include QwQ-32B (Team, 2025b), the backbone model used in the original Soft Thinking study (Zhang et al., 2025), as a reference point. This set of models spans a wide range of parameter scales (from 1.5B to 32B) and covers different weight-tying paradigms.

Benchmarks and Datasets. We use two types of datasets to evaluate different capabilities:

- Single-Token Wordlist: To test the embedding

arithmetic, we constructed a controlled vocabulary spanning 8 semantic categories (e.g., Animals, Colors, Verbs), consisting of single-token words.

- **Reasoning Benchmarks:** To evaluate the downstream performance of Soft Thinking, we verify on GSM8K (Cobbe et al., 2021), which is a dataset containing 1319 math problems with numerical answers and AIME 2024, which is a dataset containing 30 math problems with numerical answers, consistent with the evaluation suite in (Zhang et al., 2025).

Evaluation metrics. To empirically test the superposition assumption, we adopt a simple *Linearity Test*. Given a set of k tokens $\{t_1, \dots, t_k\}$ with embeddings $E(t_i)$, we construct a mixed embedding

$$v_{\text{mix}} = \frac{1}{k} \sum_{i=1}^k E(t_i),$$

and project it through the unembedding matrix W_{out} to obtain a logit vector

$$l_{\text{mix}} = W_{\text{out}} v_{\text{mix}}.$$

We report **Recall@K**, defined as the fraction of original component tokens $\{t_1, \dots, t_k\}$ that appear in the top- K logits. For the Soft Thinking reasoning experiments on GSM8K (Cobbe et al., 2021) and AIME 2024, we use the standard **Pass@1** accuracy,

$$\text{Pass@1} = \frac{c}{n},$$

where c is the number of correctly solved problems and n is the total number of instances. This allows us to compare linear decodability at the embedding level with downstream reasoning performance.

Environment. Python, Numpy, Pandas, Pytorch, Matplotlib, Transformers, scikit-learn, Jupyter Notebook, Windows Subsystem for Linux, Docker (for setting up the Soft Thinking environment)

Platform. Experiments were conducted using a combination of Google Colab, personal GPUs and institutional GPU clusters.

4 Methods

4.1 Research Question

The goal of this project can be broken down into two main sub-objectives: firstly, to determine

whether *concept tokens* can be decoded by deconstructing the superposition or convex combination of token embeddings. Our second sub-objective is to then evaluate how using **Soft Thinking** affects the performance of models with no additional fine-tuning.

4.2 Model Types

For the first objective of this project, we used Qwen3-4B, DeepSeek-1.5B and DeepSeek-7B. Since this part required determining if decoding was even possible, there is no baseline comparison to be made as in normal circumstances models perform operations on non-superposed tokens. For the second objective we used DeepSeek-1.5B and DeepSeek-7B. For this experiment, the baseline consists of the previous models tested on the same dataset without the use of **Soft Thinking**.

5 Experiments and Results

5.1 Implementation Details

For the linearity experiments on the single-token vocabulary, we vary the number of superposed tokens $k \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20\}$. For each value of k , we randomly sample k single-token words uniformly without replacement and construct a mixed embedding by averaging their input embeddings. Decoding is performed by projecting the mixed embedding through the unembedding matrix and extracting the top- K logits with $K = 30$. We report Recall@K, defined as the fraction of the original k component tokens whose token IDs appear in the top-30 decoded logits. To reduce variance from random sampling, all results are averaged over 100 independent trials per value of k .

5.2 Results and Analysis

Linear Combination Test. We first evaluate the direct decodability of superposed vectors on the single-token vocabulary. For each model, we construct a mixed embedding v_{mix} by averaging the embeddings of k randomly sampled tokens, and decode it through the unembedding matrix W_{out} . Figure 1 reports Recall@K as a function of the number of superposed tokens k for all three models. For Qwen3-4B, most original components are preserved and their tokens reliably appear in the top- K logits even as k increases. For DeepSeek-1.5B, the recall gradually decreases with k and drops sharply once the number of superposed to-

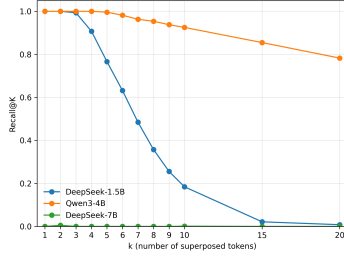


Figure 1: Linearity Test on superposed embeddings.

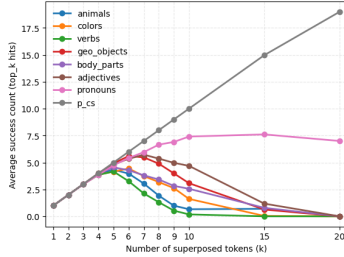


Figure 2: Average top- K hit count when mixing tokens from different semantic categories on DeepSeek-1.5B.

kens exceeds about 5. In contrast, DeepSeek-7B remains close to zero Recall@ K across the tested range of k , indicating that the mixed vector almost never decodes back to its original components. To further analyze the only model that shows non-trivial variation, we perform an additional study on DeepSeek-1.5B by grouping tokens into semantic categories. Figure 2 compares mixtures of tokens from the same category against mixtures of random tokens. Except for the high-frequency function-word category (p_cs , which contains common prepositions and conjunctions such as *of*, *in*, *on*), semantic categories behave similarly to random mixtures, suggesting that category structure offers little advantage for linear decoding beyond this special case.

Structural Geometry Analysis. To better understand the phenomena observed above, we analyze the relationship between the input and output embedding matrices of each model. For each pair (W_{in} , W_{out}), we compute the cosine similarity between the flattened matrices, together with several distance and norm statistics; the full results are summarised in Table 1. Qwen3-4B ties W_{in} and W_{out} , yielding a cosine similarity of exactly 1.0. DeepSeek-1.5B uses distinct but highly correlated matrices, with a cosine similarity of approximately 0.9658. These two models are precisely the ones that exhibit stable linear decoding in the superposition experiment. By contrast, DeepSeek-7B shows

Model	Tied	Rel. Dist.	Cos. Sim.	Norms
Qwen3-4B	Yes	0.0000	1.0000	1.09 / 1.09
DeepSeek-1.5B	No	0.2559	0.9658	1.17 / 1.14
DeepSeek-7B	No	1.3711	0.0013	1.04 / 1.03

Table 1: Relative Frobenius distance and cosine similarity between W_{in} and W_{out} , plus average row norms.

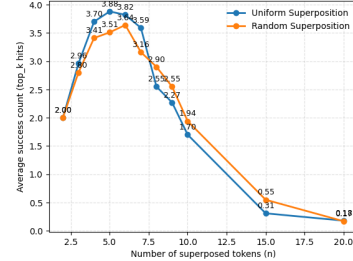


Figure 3: Average top- K hit count for non-uniform weights vs. uniform weights on DeepSeek-1.5B.

near-zero similarity (around 0.0013) despite having matching shapes and comparable embedding norms. This is exactly the model that fails the Linearity Test, consistent with the view that input and output spaces are intentionally decoupled in larger architectures.

Sensitivity to Non-Uniform Weights. Next, we repeat the decoding of superposed embeddings using non-uniform weights. For this experiment, we construct the v_{mix} mixed embedding by initializing non-uniform weights taking their softmax, then taking the weighted average of the original embeddings. This process is repeated at each iteration of the evaluation process. In Figure 3, we compare the effectiveness of decoding superpositions with non-uniform weights to those created with uniform weights. Indeed, we can observe that for smaller superpositions, uniform weights perform better than non-uniform weights, and that the opposite is true for larger superpositions, with a cross-over point at superpositions of size 7. These results are not surprising, as over a large number of trials non-uniform weights will perform similarly to uniform weights. However, the slight differences in performance can be explained by the fact that, in smaller superpositions, one token may receive a very small weight, and thus will not be present in the top- K after decoding. Alternatively, in the case of large superpositions, one token may receive a large weight, and thus easily decode back to the vocabulary space, whereas it would not have been present in the top- K tokens otherwise.

Dataset	Model	Pass@1	Avg. len (all)	Avg. len (correct)
GSM8K	DeepSeek-1.5B (no Soft Thinking)	0.7649	781	590
GSM8K	DeepSeek-7B (no Soft Thinking)	0.916	.	.
AIME 2024	DeepSeek-1.5B (no Soft Thinking)	0.3333	17138	7597
AIME 2024	DeepSeek-7B (no Soft Thinking)	0.555	.	.
GSM8K	DeepSeek-1.5B	0.7657	1357	1004
GSM8K	DeepSeek-7B	0.9022	1492	1318
AIME 2024	DeepSeek-1.5B	0.2667	15535	6254
AIME 2024	DeepSeek-7B	0.4667	11630	7328

Table 2: Soft Thinking performance on GSM8K and AIME 2024. Pass@1 and average reasoning length (tokens per problem) for DeepSeek-R1-Distill-Qwen-1.5B and 7B with and without our Soft Thinking implementation.

Soft Thinking reasoning performance. We then evaluate Soft Thinking end-to-end on DeepSeek-1.5B and DeepSeek-7B using GSM8K and AIME 2024. Table 2 reports Pass@1 and average reasoning length with and without **Soft Thinking** (DeepSeek-AI, 2025), (LLM-stats.com). On GSM8K, the 1.5B model achieves a Pass@1 of 0.77 (0.7657), with an average reasoning length of about 1357 tokens over all examples and roughly 1004 tokens on correctly solved problems. The 7B model reaches a higher Pass@1 of 0.90 (0.9022), with an average of about 1492 tokens overall and 1318 tokens on correct solutions. On AIME 2024, the 1.5B model obtains a Pass@1 of 0.27 (0.2667), with very long traces on incorrect problems; the 7B model improves this to 0.47 (0.4667), while also producing shorter average chains. These results confirm that, despite its failure in the linear superposition probe, DeepSeek-7B remains a strong Soft Thinking reasoner on both benchmarks.

Analysis. Taken together, these results reveal a clear pattern. Linear superposition is only cleanly decodable in models where the embedding and unembedding matrices are identical or highly aligned; in large untied architectures, where these matrices are nearly orthogonal, simple linear superposition fails. However, even in the presence of such linear failure, Soft Thinking built on DeepSeek-7B still achieves strong Pass@1 scores on GSM8K and non-trivial accuracy on AIME. This suggests that, for modern large models, continuous reasoning does not rely solely on naive linear operations at the input layer. Instead, the Transformer layers play a crucial role in non-linearly realigning superposed representations so that the correct answers remain decodable at the output layer. A more detailed discussion of these implications is provided in Section 6.

6 Discussion

6.1 Re-evaluating the Hypothesis

Our original proposal hypothesized that semantic superposition implies linear decodability—that convex combinations in the input space would linearly map to mixed distributions in the output. Our results falsify this “naive” assumption for modern large-scale models. While the hypothesis holds for architectures with **Weight Tying** (e.g., Qwen3-4B) due to their shared geometric basis, it collapses in **Untied** architectures (DeepSeek-7B). The structural orthogonality between input and output matrices in the 7B model prevents simple linear projection from recovering semantic concepts. This suggests that the “linear arithmetic” view of Continuous CoT may be an oversimplification that does not fully account for the geometric decoupling we observe in larger untied models.

6.2 Unexpected Observations

An unexpected and critical observation is the sharp contrast in the behavior of the **DeepSeek-7B** model. In our Linear Combination Test, the model failed completely (Recall ≈ 0), with the decoded top tokens dominated by arbitrary, semantically unrelated content words. However, when evaluated on reasoning benchmarks (AIME, GSM8K) using the full Soft Thinking pipeline, the same model achieved competitive performance. This paradox suggests that the information within the superposed embeddings is not “lost,” but rather “misaligned.” It highlights the indispensable role of the **Transformer layers**. In untied architectures, the model’s internal layers function as a necessary **non-linear projector**, rotating and aligning the input manifold (where concepts are mixed) to the orthogonal output manifold (where probabilities are defined). This implies that “Continuous CoT” in large models is not merely a process of maintaining semantic superposition, but a dynamic process of **geometric**

410 **alignment** performed by the deep network.

411 6.3 Limitations

412 A primary limitation of our study lies in the in-
413 completeness of the Soft Thinking evaluation on
414 the **Qwen3-4B** model. Due to unresolved software
415 environment conflicts (incompatibility between the
416 specific model version and the inference frame-
417 work), we were unable to obtain Pass@1 scores
418 for this baseline. This missing data point prevents
419 us from quantitatively analyzing how the output-
420 layer parameterization (tied vs. untied) affects
421 the marginal gain from Soft Thinking. Ideally, to
422 strictly isolate the effect of architecture from pa-
423 rameter size, future experiments should evaluate
424 models of identical parameter scale (e.g., two 7B
425 models) where one uses weight tying and the other
426 does not. Furthermore, our study relied on uni-
427 form weights for superposition; it remains an open
428 question whether learning task-specific mixing co-
429 efficients could mitigate the orthogonality gap in
430 untied models.

431 6.4 Future Directions

432 Based on our findings, we propose two key direc-
433 tions for future research, revisiting the secondary
434 objectives (RQ2) of our original proposal:

- 435 • **Mechanistic Interpretability of Alignment:**
436 Since the linear hypothesis fails for raw con-
437 vex combinations at the input embedding
438 layer, but decodability from the final Continu-
439 ous CoT states remains effective, the align-
440 ment must happen dynamically within the
441 model. Future work should apply probing
442 techniques (e.g., Logit Lens) to trace the evo-
443 lution of the mixed embedding layer-by-layer.
444 This would pinpoint exactly *where* and *how*
445 the Transformer resolves the orthogonality be-
446 tween input and output spaces.
- 447 • **Decodability and Similarity Analysis:** We
448 aim to return to our original goal of analyzing
449 whether the hidden states of Continuous CoT
450 are interpretable and similar to discrete CoT.
451 Specifically, we propose training a lightweight
452 **Linear Adapter** or **Projector** to map the in-
453 termediate continuous states of untied models
454 back to the vocabulary space. This would al-
455 low us to “translate” the implicit reasoning
456 steps of large models into human-readable
457 text, verifying if they follow similar logic

paths as traditional natural language reason-
ing.

7 Conclusion

In this project, our goal was to explore the feasibil-
ity of **Soft Thinking**, starting with the decompo-
sition of *concept tokens* back into the vocabulary
space, and ending by evaluating how using **Soft
Thinking** affects model performance on various
benchmarks. However, we discovered that simply
using embedding and un-embedding layers is some-
times insufficient for models that do not use **Weight
Tying**. Additionally, we observed that models that
performed poorly in the first experiment, such as
DeepSeek-7B, performed better in the benchmarks
than the other models when the full **Soft Think-
ing** pipeline was used. In future work, exploring
this disconnect could yield enlightening results that
could shed light on whether or not **Soft Thinking**
encourages models to reason in terms of concepts
rather than words, or if there is another explanation
for the gain in performance.

Finally, all code and datasets used in this work
are available at our GitHub repository.¹

8 Team Contributions

- Benjamin: Contributed to the *Linearity Test*
code and performed the *Non-Uniform Weights*
experiment. Conducted the Soft Thinking
baseline experiments on GSM8K and AIME
2024 datasets using the DeepSeek-R1-Distill-
Qwen-1.5B model. Wrote the Methods,
Conclusion and Sensitivity to Non-Uniform
Weights sections of the report. Presented the
Approach and Method, Conclusion and Exper-
iment 4 sections of the presentation.
- Edmand: Set up the GitHub repository and
implemented various functions. Created the
Single-Token Wordlist dataset and performed
the *Linearity Test* on the three LLM models us-
ing this dataset. Conducted the Soft-Thinking
experiment on the GSM8K and AIME 2024
dataset using the DeepSeek-R1-Distill-Qwen-
1.5B model. Wrote the abstract, introduction,
related work and some parts of the Data and
Environment section. Prepared and presented
the Problem Statement part of the presenta-
tion.

¹[https://github.com/edmand88/
superposition-in-continuous-chain-of-thought](https://github.com/edmand88/superposition-in-continuous-chain-of-thought)

- Yuchen: Extended the *Linearity Test* code and produced cross-model visualizations for all three LLMs. Performed the *Structural Geometry Analysis* of the input and output embedding matrices (weight tying, Frobenius distance, cosine similarity, and norm statistics). Conducted the Soft Thinking experiments on GSM8K and AIME 2024 using the DeepSeek-R1-Distill-Qwen-7B model. Writing and analysing the *Experiments and Results, Discussion* and some parts of the *Data and Environment* section, including identifying limitations and proposing future research directions. Presented first 3 experiments in the presentation.
- Michael: Provided guidance on experimental design and methodology, gave feedback, met with the students multiple times via Zoom, and proofread both the project proposal and the final report.

et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T Kwok, Zhengguo Li, Adrian Weller, and Weiyang Liu. 2023. Metamath: Bootstrap your own mathematical questions for large language models. *arXiv preprint arXiv:2309.12284*.

Zhen Zhang, Xuehai He, Weixiang Yan, Ao Shen, Chenyang Zhao, Shuohang Wang, Yelong Shen, and Xin Eric Wang. 2025. Soft thinking: Unlocking the reasoning potential of llms in continuous concept space. *arXiv preprint arXiv:2505.15778*.

Hanlin Zhu, Shibo Hao, Zhiting Hu, Jiantao Jiao, Stuart Russell, and Yuandong Tian. 2025. Reasoning by superposition: A theoretical perspective on chain of continuous thought. *arXiv preprint arXiv:2505.12514*.

References

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

DeepSeek-AI. 2025. [Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning](#).

Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. 2024. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*.

LLM-stats.com. [Gsm8k leaderboard](#).

Jacob Pfau, William Merrill, and Samuel R Bowman. 2024. Let’s think dot by dot: Hidden computation in transformer language models. *arXiv preprint arXiv:2404.15758*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Qwen Team. 2025a. [Qwen3 technical report](#).

Qwen Team. 2025b. [Qwq-32b: Embracing the power of reinforcement learning](#).

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou,